



Improving Financial Volatility Forecasting with Shape-Constrained Regularization

- A Comparative Analysis with Traditional Regularization Techniques

Nischal Adhikari

Institution: Pennsylvania State University

Motivation & Context

- **Objective:** Develop more reliable & interpretable volatility forecasts
 - Comparing traditional regularization methods with shape-constrained approaches
 - Noisy and high-volatility market conditions.
- **Hypothesis:** Shape-constrained regularization will improve forecast stability and accuracy for volatility and risk metrics compared to traditional methods.
- **Key Focus:** Implementing monotonicity constraints
 - Create structurally consistent predictions that align with expected market trends
 - Reducing noise sensitivity that often leads to misleading forecasts

Literature Review

- **Traditional Regularization:**

- Ridge, Lasso, and Elastic Net focus on reducing overfitting by penalizing large coefficients.
- Commonly applied in financial time series to handle noise and high-dimensionality.

- **Shape-Constrained Regularization:**

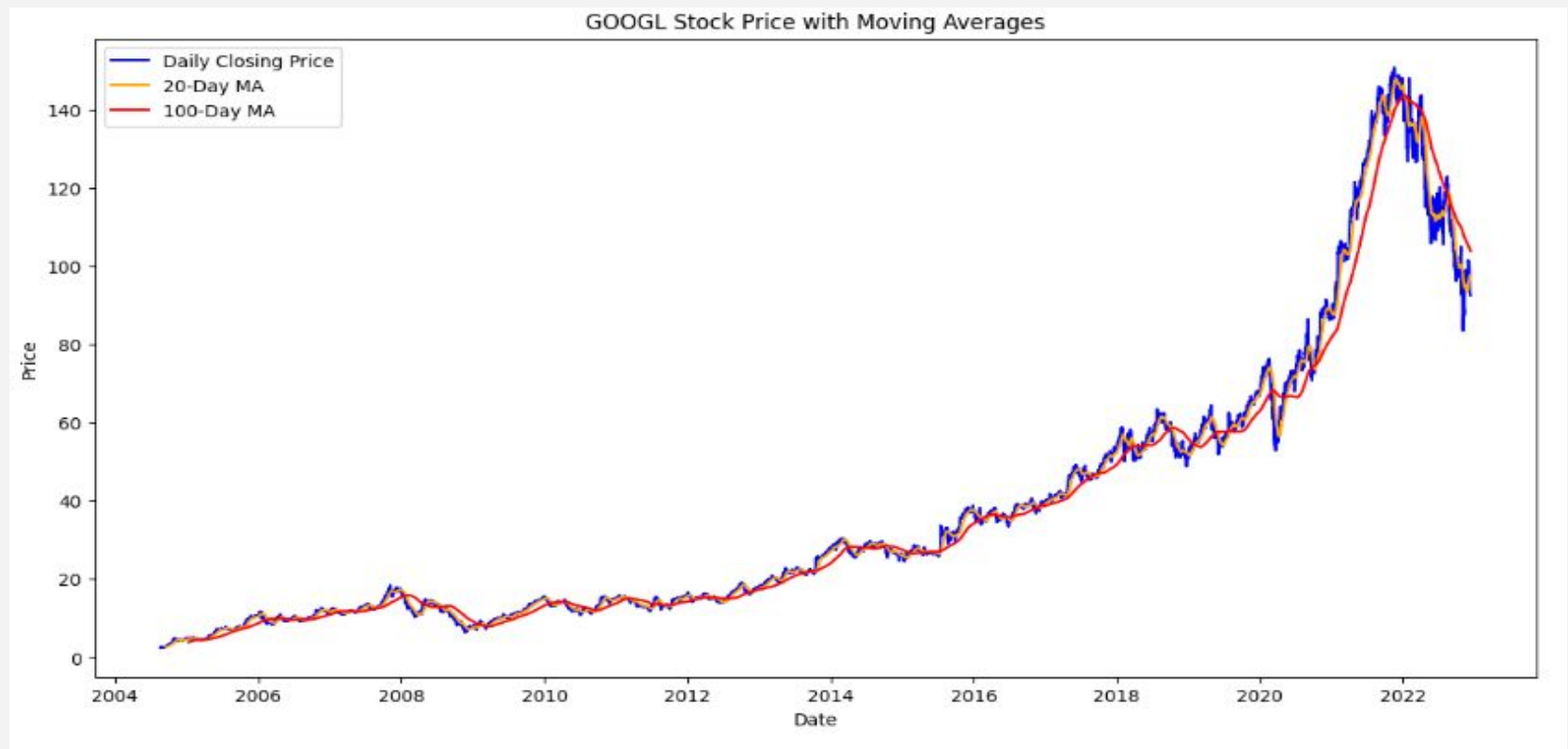
- Extends regularization by enforcing specific shapes (e.g., monotonicity).
- Has seen success in other fields (economics, engineering) but less commonly applied in finance.

- **Challenges in High-Volatility Financial Markets:**

- Market volatility and noise make accurate forecasting difficult; traditional methods often lack structural alignment with expected market patterns.

Data & Exploratory Data Analysis (EDA)

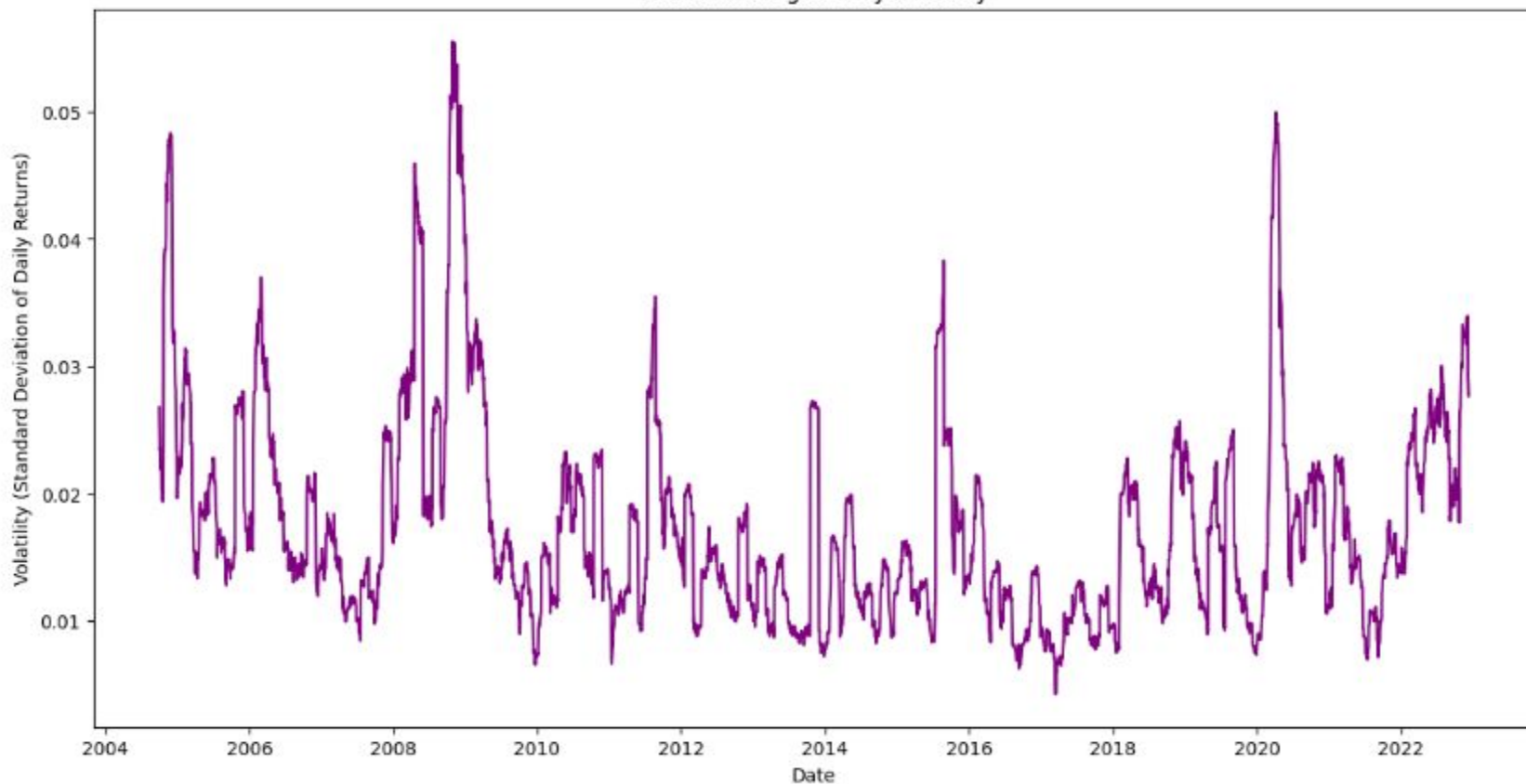
- **Data Source:** 'GOOGL' daily stock prices (Close, High, Low, Open, Volume).
- **EDA Objective:** Identify short-term and long-term trends, assess volatility patterns, and examine correlations.
- **Trend Analysis:** Moving averages to highlight short and long-term trends.



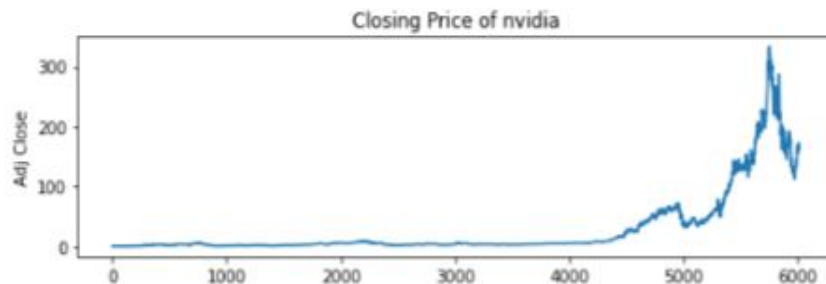
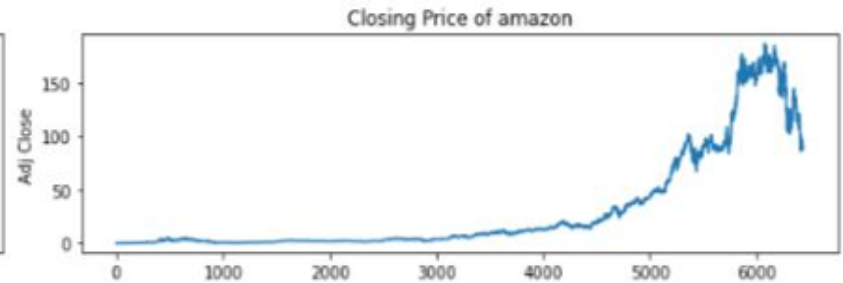
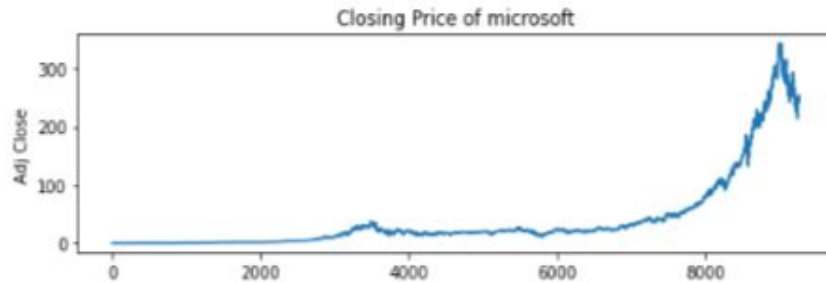
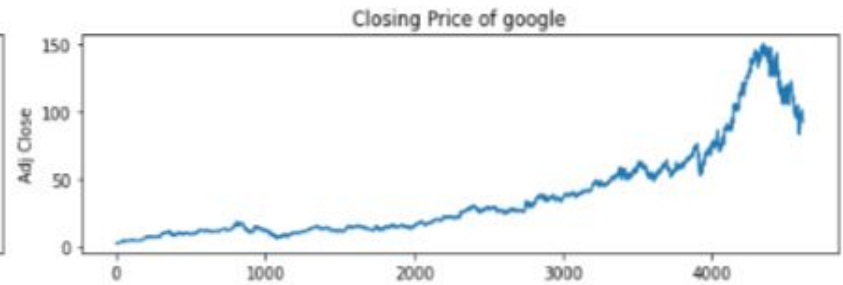
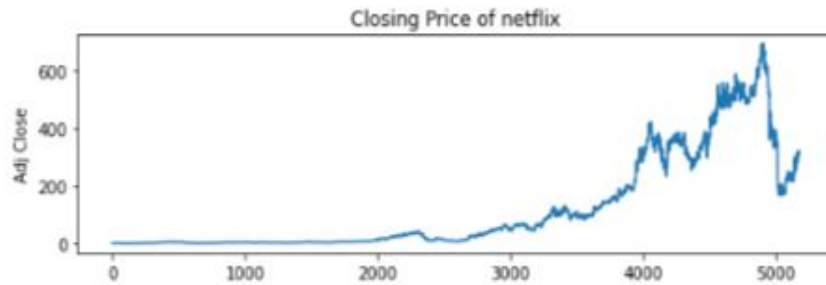
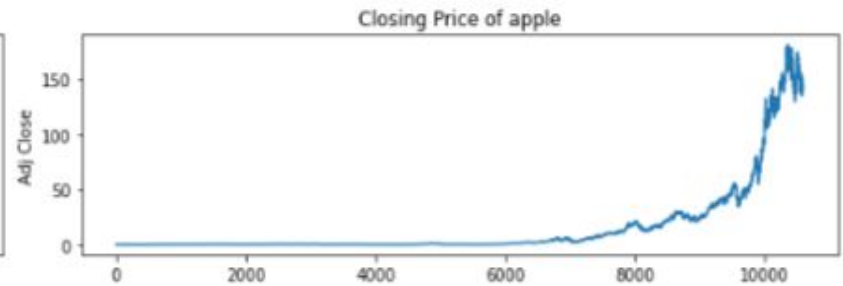
Data & Exploratory Data Analysis (EDA)

- **Volatility Assessment:**
 - Rolling standard deviation of GOOGL's daily returns over 30 days.

GOOGL Rolling 30-Day Volatility



Data & Exploratory Data Analysis (EDA)



Methodology - Traditional Regularization

- **Methods Used:** Ridge, Lasso, and Elastic Net regression models.
- **Objective:** Minimize error in volatility forecasts by penalizing large coefficients, using cross-validation to optimize hyperparameters.
- **Expected Outcome:** Stable forecasts with some sensitivity to market noise, with Ridge likely performing best among traditional methods.
- **Key Parameters:** L2 penalty for Ridge, L1 for Lasso, and a combination for Elastic Net.

Methodology - Traditional Regularization

Ridge Regression

```
ridge = GridSearchCV(Ridge(), ridge_params, cv=5, scoring='neg_root_mean_squared_error')  
ridge.fit(X_train, y_train)  
ridge_best = ridge.best_estimator_
```

Lasso Regression

```
lasso = GridSearchCV(Lasso(), lasso_params, cv=5, scoring='neg_root_mean_squared_error')  
lasso.fit(X_train, y_train)  
lasso_best = lasso.best_estimator_
```

Elastic Net Regression

```
elasticnet = GridSearchCV(ElasticNet(), elasticnet_params, cv=5, scoring='neg_root_mean_squared_error')  
elasticnet.fit(X_train, y_train)  
elasticnet_best = elasticnet.best_estimator_
```

Methodology - Shape-Constrained Regularization

- **Method:** Convex optimization using cvxpy with a monotonic constraint.
- **Objective Function:** Minimize the sum of squared differences between forecasted and actual volatility.
- **Constraints:** Monotonic (non-decreasing) constraint to smooth out volatility and reduce overreaction to random market fluctuations.
- **Expected Outcome:** More stable and smooth forecasts that align with market volatility trends, providing better interpretability in high-noise conditions.

```
# Define the objective function (minimize RMSE with monotonic constraint)
objective = cp.Minimize(cp.sum_squares(volatility_pred - y_shape))

# Define constraints for convexity (or monotonicity)
constraints = [volatility_pred[i] <= volatility_pred[i+1] for i in range(n-1)]
```

Results - Comparative Analysis

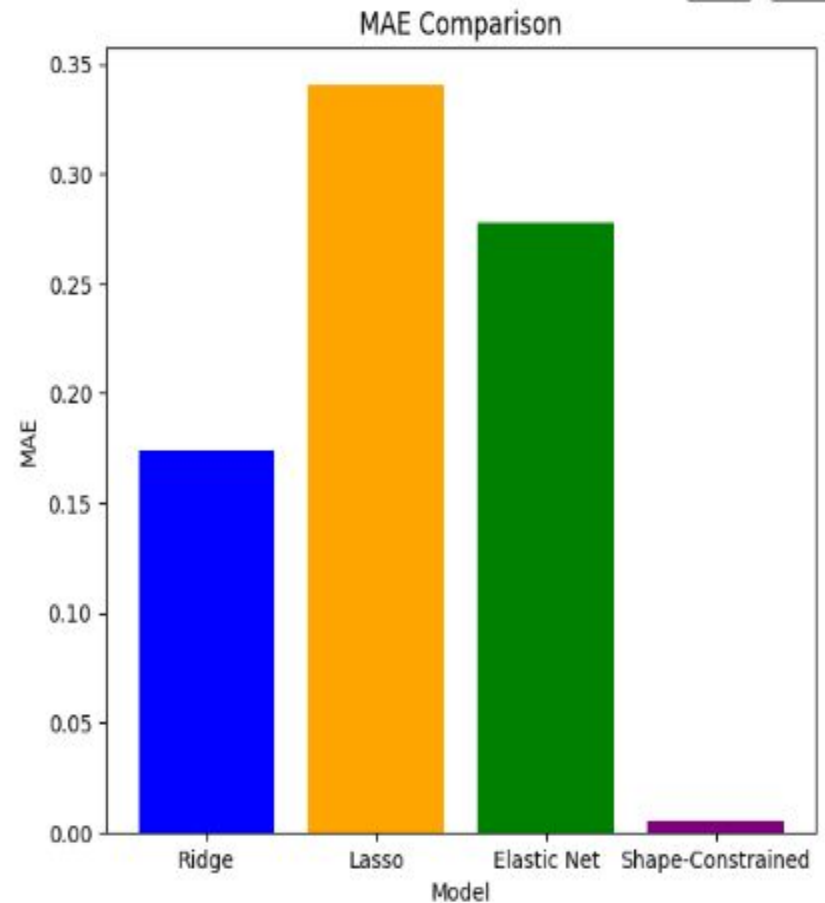
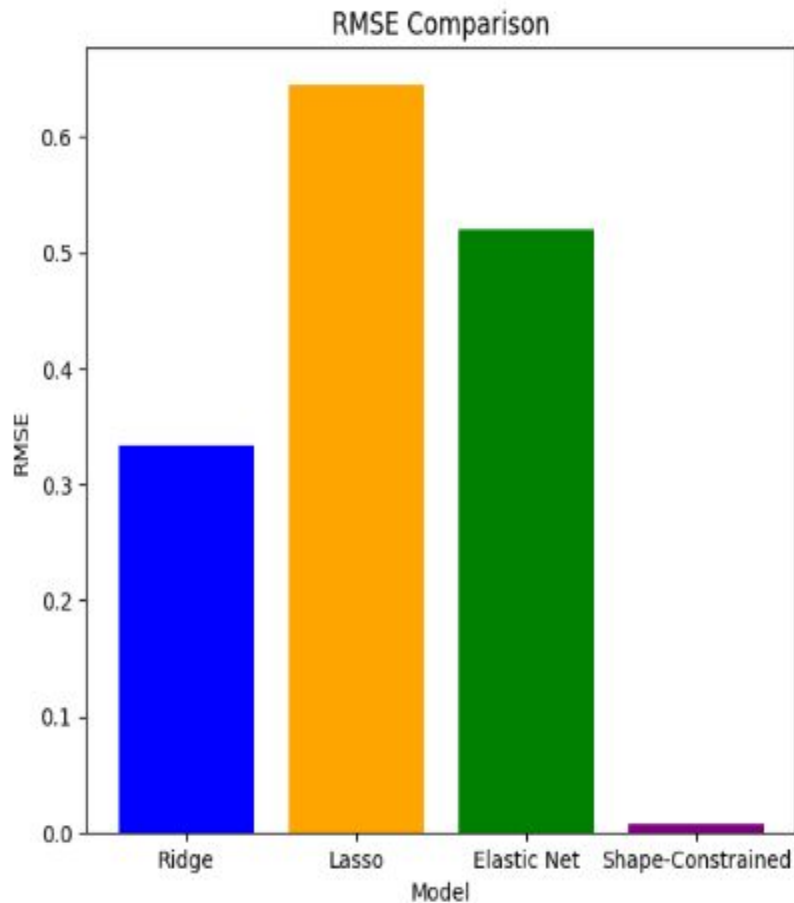
- **Metrics for Evaluation:**
 - Root Mean Squared Error (RMSE) & Mean Absolute Error (MAE) for comparing forecast accuracy and stability.
- Significantly lower RMSE & MAE for the shape-constrained model compared to traditional methods.

Comparison of Traditional and Shape-Constrained Regularization:

	Model	RMSE	MAE
0	Ridge	0.333140	0.173877
1	Lasso	0.645066	0.340578
2	Elastic Net	0.519314	0.277819
3	Shape-Constrained	0.006824	0.004896
4	Shape-Constrained	0.006824	0.004896

Results - Comparative Analysis

Comparison of Traditional and Shape-Constrained Regularization Methods



Analysis & Interpretation

- **Key Findings:**

- Shape-constrained regularization outperforms traditional methods in terms of both stability and accuracy.
- Monotonic constraints provided smoother, more interpretable forecasts, making the model less sensitive to random noise.

- **Theoretical Implications:**

- Traditional methods manage overfitting via coefficient size but lack structural alignment.
- Shape constraints align forecasts with theoretical market behavior.

- **Practical Implications:**

- Shape-constrained models offer improved tools for risk management, particularly in high-volatility scenarios.
- Potential applications in predicting volatility, VaR, and ES in real-world finance settings.

Future Directions

- Future Research:
 - Explore convexity constraints for mean-reverting volatility models.
 - Apply shape constraints to additional financial metrics (e.g., VaR and ES) to validate generalizability.
 - Test & forecast on other datasets (stocks).
- Open Questions:
 - How might shape constraints interact with real-time, high-frequency data?
 - Could these models be further improved by integrating additional market indicators?

References

- fatimamansoor28. (2024, April 23). Performing EDA on stock market data. Kaggle.
<https://www.kaggle.com/code/fatimamansoor28/performing-eda-on-stock-market-data>
- Mooney, P. (2023, January 16). Stock market data (NASDAQ, NYSE, S&P500). Kaggle.
<https://www.kaggle.com/datasets/paultimothymooney/stock-market-data>
- Nur, A. R., Jaya, A. K., & Siswanto, S. (2023). Comparative analysis of Ridge, lasso, and elastic net regularization approaches in handling multicollinearity for infant mortality data in South Sulawesi. Jurnal Matematika, Statistika Dan Komputasi, 20(2), 311–319. <https://doi.org/10.20956/j.v20i2.31632>
- Y. Liang, A. Thavaneswaran, Z. Zhu, R. K. Thulasiram and M. E. Hoque, "Data-Driven Adaptive Regularized Risk Forecasting," 2020 IEEE 44th Annual Computers, Software, and Applications Conference (COMPSAC), Madrid, Spain, 2020, pp. 1296-1301, doi: 10.1109/COMPSAC48688.2020.00-77. keywords: {Predictive models;Reactive power;Forecasting;Correlation;Computational modeling;Adaptation models;Standards;Regularized Adaptive Forecasts, Data-Driven Volatility Model, EWMA, VaR, ES, Model Risk Ratio, Elastic Net, Probabilistic Forecasts},